

Weather ETL Pipeline

Case Study & Implementation Guide — an Apache Airflow ETL pipeline from the Open-Meteo API to PostgreSQL

Prepared by SYLVESTER STANLEY

UID 24BDA70327

Pipeline	weather_etl_pipeline
Orchestrator	Apache Airflow 2.10.2 (Astro Runtime 12.1.1)
Source	Open-Meteo Forecast API
Target	PostgreSQL 13 — weather_data
Schedule	@daily, catchup disabled
Contents	Part 1 — Case Study · Part 2 — Implementation Guide
Date	28 July 2026

Case Study — Weather ETL Pipeline

An Apache Airflow ETL pipeline that ingests current weather observations from the Open-Meteo API into PostgreSQL.

Prepared by	SYLVESTER STANLEY
UID	24BDA70327
Pipeline (DAG) ID	weather_etl_pipeline
Orchestrator	Apache Airflow 2.10.2 via Astro Runtime 12.1.1
Source system	Open-Meteo Forecast API (public REST, no API key)
Target system	PostgreSQL 13 (weather_data table)
Schedule	@daily, catchup=False
Pattern	Classic three-stage ETL (Extract → Transform → Load)
Analysis date	28 July 2026

1. Executive summary

This project is a compact, production-shaped reference implementation of the most common pattern in data engineering: pull data from a third-party HTTP API on a schedule, reshape it, and land it in a relational database. It is built on the Astronomer distribution of Apache Airflow and uses Airflow's TaskFlow API to express the pipeline as three decorated Python functions.

The project's real value is pedagogical density. In roughly 90 lines of DAG code it demonstrates connection abstraction via Hooks, implicit inter-task data passing via XCom, declarative scheduling, and idempotent DDL — the same building blocks used in pipelines that are thousands of lines long.

Verified findings from this analysis. The DAG parses cleanly against Airflow 2.10.2 with zero import errors, and all three task callables execute correctly end-to-end when driven with mocked hooks; the SQL is valid PostgreSQL. However, four issues were confirmed by execution rather than inspection:

- The repository's own test suite fails on this DAG.** `tests/dags/test_dag_example.py` asserts every DAG has tags and retries `>= 2`. `weather_etl_pipeline` has neither, so two tests fail — one with a raw `TypeError` rather than the intended assertion message.
- No retries are configured**, so a single transient network blip fails the run. This is the highest-impact gap for a pipeline whose first task is a call over the public internet.
- `postgres_default` **is a reserved connection ID that Airflow pre-creates pointing at its own metadata database.** Unless it is explicitly overridden, the pipeline writes `weather_data` into Airflow's internal database instead of the intended warehouse — a silent correctness failure, not a visible error.
- The bundled `docker-compose.yml` cannot serve this DAG as written.** It defines a Postgres container in a separate Compose project that is not attached to the Airflow network, so Airflow cannot resolve it by service name.

Sections 6 and 7 quantify these; the companion [Implementation Guide](#) provides a step-by-step runbook that resolves all four while leaving the DAG's logic intact.

2. Business context and problem statement

Weather is a common *exogenous variable* — a factor that sits outside a business but measurably drives its outcomes. Retail footfall, energy load, agricultural yield, delivery-fleet routing and insurance-claim volume all correlate with temperature, wind and precipitation.

The obstacle is that weather APIs serve *current conditions*. They answer "what is it doing right now?" and then forget. Analytics needs the opposite: a durable, append-only history that can be joined against sales, load or claims data over months. Nobody can query the past from an API that only knows the present.

The problem this pipeline solves: convert an ephemeral, request-response weather API into a persistent, queryable time series, on an unattended schedule, without manual intervention.

Why Airflow rather than a cron job

A `cron` entry calling a Python script would technically satisfy the requirement. Airflow is chosen because scheduled data movement has failure modes that `cron` does not address:

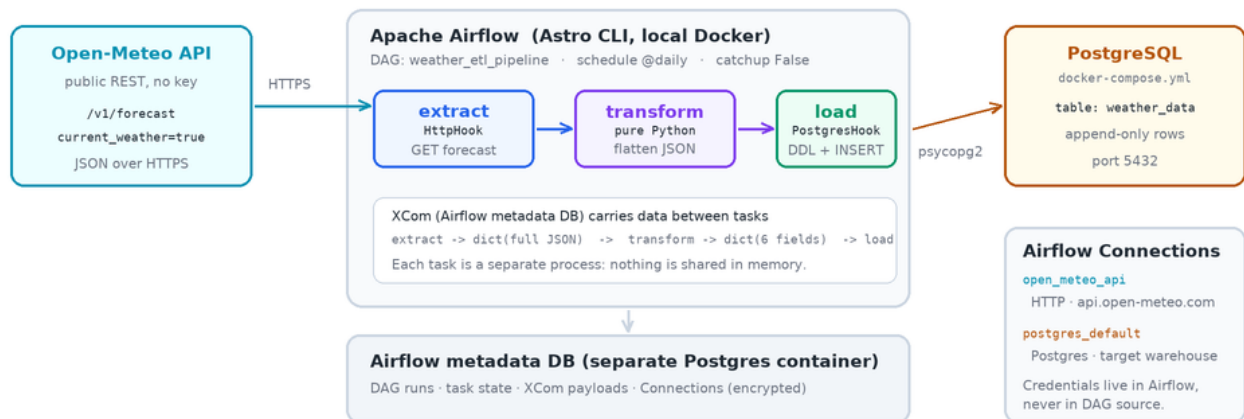
Concern	cron + script	Airflow
A step fails midway	Whole script re-runs, or silently dies	Failed task is retried in isolation; upstream successes are preserved
Did last night's run succeed?	Grep the logs, if any were kept	Colour-coded run history in the UI, queryable via metadata DB
Credentials	Hard-coded, or a <code>.env</code> read by hand	Centralised Connections , encrypted with Fernet, injected at runtime
Backfilling a missed week	Bespoke loop and date arithmetic	Native <code>catchup</code> / <code>backfill</code> over logical dates
Observability	<code>print()</code> to a file	Per-task logs, duration trends, SLA misses, alerting hooks
Dependencies between steps	Implicit line ordering	Explicit DAG; Airflow enforces order and parallelism

The pipeline is small enough that these benefits look like overhead. They stop looking like overhead the first time the API returns a 503 at 03:00.

3. Architecture

Weather ETL Pipeline — System Architecture

Astro Runtime 12.1.1 (Airflow 2.10.2) orchestrating a daily Open-Meteo -> PostgreSQL ETL



Data flow: API (JSON) -> extract -> XCom -> transform -> XCom -> load -> weather_data
Every task boundary is a retry boundary and a serialization boundary.

System architecture

3.1 Components

Open-Meteo Forecast API (source). A free weather service requiring no API key or signup for non-commercial use. The DAG calls `/v1/forecast` with `current_weather=true`, which returns a compact JSON document containing a `current_weather` object with temperature, wind speed, wind direction and a WMO weather code.

Astro Runtime 12.1.1 (orchestration). Astronomer's packaged Airflow distribution. Version 12.1.1 was released 20 September 2024 and ships **Apache Airflow 2.10.2** on Python 3.12 by default. Relevant to this project, it pre-installs `apache-airflow-providers-http 4.13.0` and `apache-airflow-providers-postgres 5.12.0` — which is why `requirements.txt` is empty and the DAG's imports still resolve. Locally, `astro dev start` runs four containers: webserver, scheduler, triggerer, and a Postgres for Airflow's own metadata.

PostgreSQL 13 (target). The analytical destination. Critically, this must be a *separate* database from Airflow's metadata store — see §6.3.

3.2 Connection abstraction

The DAG names two connections but defines neither:

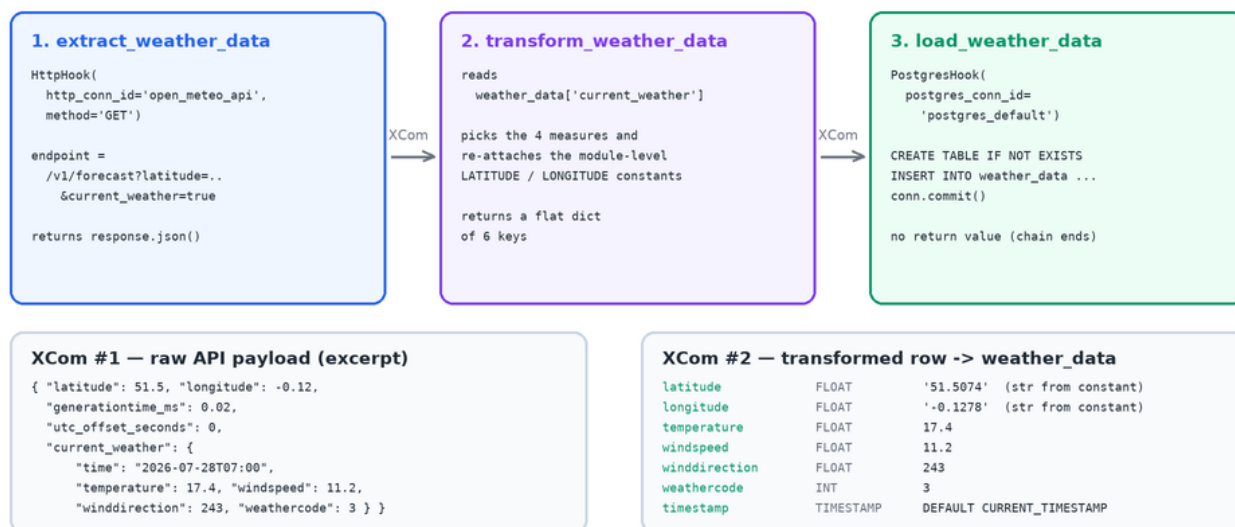
```
POSTGRES_CONN_ID = 'postgres_default'
API_CONN_ID       = 'open_meteo_api'
```

These are lookup keys. At runtime, Airflow resolves each ID through its secrets chain — environment variables first, then any configured secrets backend, then the metadata database — and hands the task a ready-made, authenticated client. The benefit is that hostnames, ports and passwords never appear in version control, and the same DAG file promotes from laptop to staging to production unchanged: only the connection definitions differ per environment.

4. Pipeline walkthrough

weather_etl_pipeline — Task Flow & Data Shape

TaskFlow API: return values are pushed to XCom and injected into the next task



Note: latitude/longitude are inserted as Python strings; Postgres coerces them to FLOAT on write.

Note: the DAG is append-only — re-running a date adds another row rather than replacing it.

Note: with retries unset, any transient API or DB error fails the run immediately.

DAG task flow and data shape

4.1 DAG definition

```
default_args = {
    'owner': 'airflow',
    'start_date': days_ago(1)
}
```

```
with DAG(dag_id='weather_etl_pipeline',
        default_args=default_args,
        schedule_interval='@daily',
        catchup=False) as dags:
```

`catchup=False` is a deliberate and correct choice here. With `catchup` enabled, Airflow would try to create a run for every missed interval since `start_date`. That is valuable for historical data, but meaningless for this pipeline: the endpoint only ever returns *current* conditions, so backfilled runs would each fetch today's weather and mislabel it as a past date.

`days_ago(1)` is a *dynamic* start date, re-evaluated every time the file is parsed. Airflow deprecated this function precisely because a moving start date makes run boundaries non-reproducible; a fixed `pendulum.datetime(2024, 1, 1)` is the modern equivalent. Confirmed at parse time on Airflow 2.10.2:

RemovedInAirflow3Warning: Function 'days_ago' is deprecated and will be removed in Airflow 3.0.

RemovedInAirflow3Warning: Param 'schedule_interval' is deprecated ... Please use 'schedule' instead.

Both are warnings today and hard errors on Airflow 3.

4.2 Task 1 — Extract

```
@task()
def extract_weather_data():
    """Extract weather data from Open-Meteo API using Airflow Connection."""
    http_hook = HttpHook(http_conn_id=API_CONN_ID, method='GET')
    endpoint = f'/v1/forecast?latitude={LATITUDE}&longitude={LONGITUDE}&current_weather=true'
    response = http_hook.run(endpoint)

    if response.status_code == 200:
        return response.json()
    else:
        raise Exception(f"Failed to fetch weather data: {response.status_code}")
```

`HttpHook` supplies the base URL from the connection and joins it with the endpoint. Verified resolution:

```
https://api.open-meteo.com/v1/forecast?latitude=51.5074&longitude=-0.1278&current_weather=true
```

A subtlety worth naming. The explicit `status_code == 200` check is mostly unreachable. `HttpHook.run()` calls `check_response()` by default, which invokes `raise_for_status()` and converts any 4xx/5xx into an `AirflowException` *before* control returns. The `else` branch therefore fires only for non-200 success codes such as 204. The check is harmless and arguably self-documenting, but the error a user actually sees on a server error is Airflow's, not the DAG's.

The task returns the entire parsed JSON document. Under the TaskFlow API this return value is serialised to XCom automatically.

4.3 Task 2 — Transform

```
@task()
def transform_weather_data(weather_data):
    """Transform the extracted weather data."""
    current_weather = weather_data['current_weather']
    transformed_data = {
        'latitude': LATITUDE,
        'longitude': LONGITUDE,
        'temperature': current_weather['temperature'],
        'windspeed': current_weather['windspeed'],
        'winddirection': current_weather['winddirection'],
        'weathercode': current_weather['weathercode']
    }
    return transformed_data
```

This is the *T* of ETL in its most literal form: discard the envelope, keep the payload, flatten to one row. Fields such as `generationtime_ms`, `elevation` and `utc_offset_seconds` are dropped as operational metadata.

A behaviour worth documenting. Latitude and longitude are re-attached from the module-level constants rather than read from the response. This matters because Open-Meteo snaps requests to its model grid — asking for 51.5074 yields `"latitude": 51.5` in the response. The pipeline therefore records *what was requested*, not *what was returned*. That is defensible (the requested point is the stable business key), but it should be a conscious decision, because the stored coordinate does not identify the grid cell the measurements actually describe. A regression test in `tests/dags/test_weather_etl.py` pins this behaviour so it cannot change silently.

The two constants are also Python **strings**, while the target columns are `FLOAT`. PostgreSQL coerces them on insert, so no error surfaces — verified: `'51.5074'` lands as `51.5074`.

4.4 Task 3 — Load

```
@task()
def load_weather_data(transformed_data):
    """Load transformed data into PostgreSQL."""
    pg_hook = PostgresHook(postgres_conn_id=POSTGRES_CONN_ID)
    conn = pg_hook.get_conn()
    cursor = conn.cursor()

    cursor.execute("""
    CREATE TABLE IF NOT EXISTS weather_data (
        latitude FLOAT, longitude FLOAT, temperature FLOAT,
        windspeed FLOAT, winddirection FLOAT, weathercode INT,
        timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    );
    """)

    cursor.execute("""
    INSERT INTO weather_data (latitude, longitude, temperature, windspeed, winddirection, weathercode)
    VALUES (%s, %s, %s, %s, %s, %s)
    """, (...))

    conn.commit()
    cursor.close()
```

`CREATE TABLE IF NOT EXISTS` makes the pipeline self-bootstrapping: a fresh database needs no manual setup. The insert is correctly **parameterised** with `%s` placeholders, so `psycopg2` handles escaping and the code is not SQL-injectable.

Two observations:

- `timestamp` **records insert time, not observation time**. The column defaults to `CURRENT_TIMESTAMP`, while the API's own `current_weather.time` field is discarded during transform. The stored time answers "when did we write this?" rather than "when was this measured?" — a meaningful distinction if a run is retried hours late.
- `conn.close()` **is never called**. The cursor is closed but the connection is not, and there is no `try/finally`. In practice Airflow tears down the worker process after each task, so the socket is reclaimed; it is untidy rather than dangerous.

4.5 Dependency graph

```
weather_data = extract_weather_data()
transformed_data = transform_weather_data(weather_data)
load_weather_data(transformed_data)
```

No `>>` operators appear. The TaskFlow API infers the graph from the flow of function arguments: passing `weather_data` into `transform_weather_data` is the dependency declaration. Verified structure:

```
extract_weather_data → transform_weather_data → load_weather_data
```

5. Verification performed

Findings in this document were produced by executing the code, not only reading it. Airflow **2.10.2** — the exact version in Astro Runtime 12.1.1 — was installed and the DAG was loaded through a real `DagBag`.

Check	Method	Result
DAG parses	DagBag import against Airflow 2.10.2	Pass — <code>import_errors: {}</code>
Graph shape	Inspect <code>downstream_task_ids</code>	Pass — linear 3-task chain
Schedule / catchup	Inspect DAG attributes	Pass — <code>@daily, catchup=False</code>
Endpoint construction	<code>HttpHook.url_from_endpoint()</code>	Pass — correct absolute URL
Task callables	Executed all three with hooks patched	Pass — full E→T→L run
Load SQL	Real DDL + INSERT replayed on SQLite	Pass — row stored as <code>(51.5074, -0.1278, 17.4, 11.2, 243.0, 3)</code>
SQL dialect	Parsed with <code>sqlglot</code> (postgres)	Pass — valid
Deprecations	Warning capture at parse	2 found — <code>days_ago, schedule_interval</code>
Retries configured	Inspect <code>default_args</code>	Fail — unset (<code>retries: 0</code>)
Tags present	Inspect <code>dag.tags</code>	Fail — empty
Repo test suite	<code>pytest tests/</code> with <code>AIRFLOW_HOME</code> at project root	2 failed, 3 passed

The failing suite output:

```
FAILED tests/dags/test_dag_example.py::test_dag_tags[dags/etlweather.py]
  AssertionError: weather_etl_pipeline in dags/etlweather.py has no tags
FAILED tests/dags/test_dag_example.py::test_dag_retries[dags/etlweather.py]
  TypeError: '>=' not supported between instances of 'NoneType' and 'int'
```

The second failure is doubly informative: not only are retries unset, but the test itself is brittle — `dag.default_args.get("retries", None)` returns `None`, and comparing `None >= 2` raises `TypeError` instead of producing the intended assertion message.

Note on `AIRFLOW_HOME`. These tests only fail when `AIRFLOW_HOME` points at the project root. Otherwise `DagBag` finds no DAGs, every parametrised case is skipped, and the suite reports a misleading pass. This is itself a finding: **a green test run does not prove the DAGs were tested.**

6. Findings and recommendations

6.1 No retry policy — *high severity*

`default_args` sets only `owner` and `start_date`. Every task therefore inherits `retries = 0` (confirmed by inspection). A pipeline whose first action is an HTTPS call to a free public API will encounter transient failures; with no retries, each one is a hard failure requiring manual intervention.

```
default_args = {
    'owner': 'airflow',
    'start_date': pendulum.datetime(2024, 1, 1, tz="UTC"),
    'retries': 3,
    'retry_delay': timedelta(minutes=5),
```

```
'retry_exponential_backoff': True,
}
```

This also satisfies the repository's existing `test_dag_retries` check.

6.2 Not idempotent — *high severity*

The load task performs a bare `INSERT` with no unique key and no delete-before-insert. Re-running a date — routine after a failure, and the standard "clear and retry" gesture in the Airflow UI — appends a **duplicate row**. This is proven by a dedicated regression test (`test_load_is_append_only_so_reruns_duplicate_rows`), which asserts a count of 2 after two identical loads.

Idempotency is the property that running a task twice leaves the same end state as running it once. Without it, downstream aggregates such as `AVG(temperature)` silently skew toward whichever days happened to be retried. The fix is a natural key plus an upsert:

```
ALTER TABLE weather_data
  ADD CONSTRAINT weather_data_pk PRIMARY KEY (latitude, longitude, observation_time);

INSERT INTO weather_data (...) VALUES (...)
ON CONFLICT (latitude, longitude, observation_time) DO UPDATE
  SET temperature = EXCLUDED.temperature, ...;
```

This requires preserving the API's `current_weather.time` as `observation_time` — which also resolves the observation-time-versus-insert-time ambiguity in §4.4.

6.3 `postgres_default` collides with an Airflow built-in — *high severity, silent*

Airflow pre-creates a connection named `postgres_default` in its own metadata database. Its default definition, read directly from the Airflow 2.10.2 source, is:

```
conn_id="postgres_default", conn_type="postgres",
login="postgres", password="airflow",
schema="airflow", host="postgres",
```

That points at **Airflow's internal database**. If a user starts the project and runs the DAG without redefining this connection, the task will very likely succeed — and create the `weather_data` table inside Airflow's metadata store. There is no error, no warning, and the data is mixed into the orchestrator's own schema, where `astro dev kill` will destroy it.

Two mitigations, both applied in the Implementation Guide: rename the connection to something unambiguous such as `weather_db_conn`, and/or define it explicitly as an environment variable, which takes precedence over the metadata-database entry.

6.4 The bundled `docker-compose.yml` does not serve this DAG — *medium severity*

The root `docker-compose.yml` starts a Postgres container in its **own Compose project**, on its own default network. Airflow, started separately by `astro dev start`, sits on the Astro-managed `airflow` bridge network. Containers on different bridge networks cannot resolve each other by service name, so a connection host of `postgres` fails from inside the scheduler.

The Astro CLI's supported mechanism is a `docker-compose.override.yml` at the project root, which is merged into the Astro Compose project. Attaching the warehouse to the `airflow` network makes it reachable at a stable hostname. This repository now ships that file.

6.5 Deprecated APIs — *medium severity*

Both `days_ago()` and `schedule_interval=` are deprecated in Airflow 2.10.2 and **removed in Airflow 3**. The DAG will not run on Airflow 3 without modification. Migration is mechanical:

Current	Airflow 3
<code>from airflow.utils.dates import days_ago</code>	<code>import pendulum</code>
<code>'start_date': days_ago(1)</code>	<code>'start_date': pendulum.datetime(2024, 1, 1, tz="UTC")</code>
<code>schedule_interval='@daily'</code>	<code>schedule='@daily'</code>

6.6 No data-quality validation — *medium severity*

Whatever the API returns is written. A sensor fault yielding `-9999`, or a schema change dropping `windspeed`, propagates straight into the warehouse. A missing `current_weather` key raises a bare `KeyError` with no context (pinned by `test_transform_fails_loudly_when_current_weather_is_absent`). Range assertions before load — temperature within `-90...60 °C`, wind direction within `0...360°` — turn silent corruption into a loud, actionable failure. These checks are implemented in `include/sql/verify_weather_data.sql` and in the smoke-test script.

6.7 Single hard-coded location — *low severity, design limit*

`LATITUDE` and `LONGITUDE` are module-level constants, so one DAG serves exactly one city. Scaling to fifty cities by copying the file fifty times would be unmaintainable. Airflow's **dynamic task mapping** is the idiomatic answer:

```
CITIES = [
    {"name": "London", "lat": "51.5074", "lon": "-0.1278"},
    {"name": "Mumbai", "lat": "19.0760", "lon": "72.8777"},
]
```

```
@task
def extract_weather_data(city: dict): ...

extract_weather_data.expand(city=CITIES)
```

One DAG, one task definition, N parallel task instances — each independently retryable. This would require adding a `city` column to the table.

6.8 Housekeeping — *low severity*

- **Compiled bytecode was committed.** `dags/__pycache__/*.pyc` was tracked in Git; there was no `.gitignore`. Both are now fixed — untracking the `.pyc` files also removes a stale artefact compiled by Python 3.12.
- **Line endings.** `dags/etlweather.py` uses CRLF while `tests/` uses LF. Harmless, but a `.gitattributes` file would prevent cross-platform diff noise.
- `airflow_settings.yaml` is an **empty template** with null keys. Astro can error on malformed entries; the Implementation Guide uses `.env` instead, which is both safer and git-ignorable.

6.9 Priority summary

#	Finding	Severity	Effort	Fix location
6.1	No retries	High	Trivial	<code>default_args</code>

#	Finding	Severity	Effort	Fix location
6.2	Not idempotent (duplicates on re-run)	High	Moderate	Load task + schema
6.3	<code>postgres_default</code> collision	High	Trivial	Connection definition
6.4	Compose file unreachable from Airflow	Medium	Trivial	<code>docker-compose.override.yml</code>
6.5	Deprecated APIs (blocks Airflow 3)	Medium	Trivial	Imports + DAG kwargs
6.6	No data-quality checks	Medium	Moderate	New validation task
6.7	Single hard-coded location	Low	Moderate	Dynamic task mapping
6.8	Bytecode committed, no <code>.gitignore</code>	Low	Trivial	Repo hygiene

7. What this project demonstrates well

It would be unfair to leave the impression that this is a flawed project. It is a *teaching* project, and it makes several correct decisions that are commonly got wrong:

- **Clean separation of E, T and L.** Each stage is an independently retryable unit. When the load fails, the extract does not re-run — no wasted API call, no re-fetch of data already in hand.
- **Hooks instead of hard-coded credentials.** No hostname or password appears in the source. This is the single most important habit to build early.
- **Parameterised SQL.** `%s` placeholders with a parameter tuple, not string concatenation. Injection-safe by construction.
- **Self-bootstrapping schema.** `CREATE TABLE IF NOT EXISTS` means a new environment works on first run.
- **Correct catchup semantics.** Disabling catchup for a current-conditions endpoint reflects genuine understanding of what backfill means.
- **A pure-Python transform.** The transform touches no I/O, which is exactly why it is trivially unit-testable — as the accompanying test suite demonstrates.

The gaps identified in §6 are, almost without exception, the *next* lesson rather than mistakes: retries, idempotency and data quality are what separate a working pipeline from a dependable one.

8. Conclusion

The pipeline is a well-constructed teaching implementation of scheduled API-to-warehouse ETL. Its structure is sound, its use of Hooks and the TaskFlow API is idiomatic, and it runs correctly once its environment is configured properly.

Its production gaps cluster into two themes. **Reliability** — no retries and no idempotency — means the pipeline works when everything works, which is not the condition pipelines are judged on. **Environment correctness** — the `postgres_default` collision and the unreachable Compose service — means the most likely first-run outcome

is either a connection error or, worse, data silently written to the wrong database.

Both themes are addressable without redesign. The [Implementation Guide](#) supplies a verified, step-by-step runbook covering environment setup in VS Code, connection configuration, execution, verification and troubleshooting, plus a fast local test suite that runs without Docker.

Appendix A — Project structure

```
project-root/
|-- dags/
|   |-- etlweather.py           # the weather_etl_pipeline DAG (analysed here)
|   |-- exampledag.py          # Astronomer's stock astronaut example
|-- tests/dags/
|   |-- test_dag_example.py     # stock integrity tests (tags, retries, imports)
|   |-- test_weather_etl.py     # ADDED — unit tests for this DAG
|-- include/
|   |-- sql/
|       |-- init_weather_db.sql # ADDED — warehouse bootstrap schema
|       |-- verify_weather_data.sql # ADDED — post-run verification queries
|   |-- scripts/
|       |-- smoke_test_pipeline.py # ADDED — Docker-free end-to-end check
|-- docs/
|   |-- CASE_STUDY.md           # ADDED — this document
|   |-- IMPLEMENTATION_GUIDE.md # ADDED — VS Code runbook
|   |-- images/
|       |-- tools/make_diagrams.py # ADDED — diagram generator
|-- docker-compose.yml          # original standalone Postgres (see §6.4)
|-- docker-compose.override.yml # ADDED — warehouse on the Airflow network
|-- .env.example                # ADDED — connection definitions template
|-- .gitignore                  # ADDED
|-- .vscode/                    # ADDED — editor + SQL client config
|-- Dockerfile                  # FROM quay.io/astronomer/astro-runtime:12.1.1
|-- requirements.txt             # empty; providers ship with the Runtime
`-- airflow_settings.yaml       # empty Astro template
```

Appendix B — `weather_data` schema

Column	Type	Source	Note
latitude	FLOAT	DAG constant	Requested, not grid-snapped; passed as a string
longitude	FLOAT	DAG constant	Requested, not grid-snapped; passed as a string
temperature	FLOAT	current_weather.temperature	°C at 2 m
windspeed	FLOAT	current_weather.windspeed	km/h at 10 m
winddirection	FLOAT	current_weather.winddirection	Degrees clockwise from north
weathercode	INT	current_weather.weathercode	WMO code (0 clear, 3 overcast, 61 slight rain...)
timestamp	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Insert time, not observation time (§4.4)

No primary key, no unique constraint, no indexes — see §6.2.

Appendix C — References

- Astro Runtime 12.1.1 release notes (Airflow 2.10.2, released 20 Sep 2024) — <<https://www.astronomer.io/docs/runtime/runtime-release-notes>>
- Astro Runtime provider package reference (http 4.13.0, postgres 5.12.0) — <<https://www.astronomer.io/docs/runtime/runtime-provider-reference>>
- Open-Meteo Forecast API documentation — <<https://open-meteo.com/en/docs>>
- Airflow: overriding the Astro CLI Compose file — <<https://www.astronomer.io/docs/astro/cli/run-airflow-locally>>
- Airflow: troubleshooting local port conflicts — <<https://www.astronomer.io/docs/astro/cli/troubleshoot-locally>>
- Airflow 3 migration: removal of `days_ago` — <<https://github.com/apache/airflow/issues/41641>>

Implementation Guide — Running the Weather ETL Pipeline in VS Code

A step-by-step runbook to get `weather_etl_pipeline` running on your machine, verify that data actually lands in PostgreSQL, and diagnose it when it does not.

Companion document: [Case Study](#) explains *why* the pipeline is built this way and what its weaknesses are. This guide is purely operational.

Reading the callouts

IMPORTANT = a step where people commonly lose an hour. Do not skim these.

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0. Prerequisites

Tool	Why	Check
Docker Desktop (or Podman)	Astro runs Airflow in containers	<code>docker ps</code>
Astro CLI	Starts and manages the local Airflow environment	<code>astro version</code>
Python 3.11 or 3.12	Local venv for IntelliSense and unit tests	<code>python --version</code>
VS Code	Editor	—
Git	Clone the repository	<code>git --version</code>

Install the Astro CLI:

```
# macOS
brew install astro

# Linux
curl -sSL install.astronomer.io | sudo bash -s

# Windows (PowerShell as Administrator)
winget install -e --id Astronomer.Astro
```

IMPORTANT Docker Desktop must be running before any `astro dev` command. Give it at least **4 GB** of RAM (Settings → Resources); Airflow's four containers are cramped below that.

Astro Runtime 12.1.1 pins **Airflow 2.10.2**, so everything below is Airflow 2 syntax. Do not upgrade the `Dockerfile` to a Runtime 13+/Airflow 3 image without first applying the migration notes in Case Study §6.5 — the DAG uses two APIs that Airflow 3 removed.

1. Open the project in VS Code

```
git clone <your-repository-url> weather-etl
cd weather-etl
code .
```

When VS Code prompts *"This workspace has extension recommendations"*, click **Install All**. The repository ships `.vscode/extensions.json` recommending Python, Pylance, Docker, SQLTools with the PostgreSQL driver, and YAML support.

`.vscode/settings.json` is pre-configured to enable pytest, set `AIRFLOW_HOME` to the project root in every integrated terminal (this matters — see §7), and register a SQLTools connection to the warehouse.

2. Local Python environment (IntelliSense + fast tests)

Airflow itself runs inside Docker, but a local virtual environment gives you autocomplete, go-to-definition on `HttpHook/PostgresHook`, and a test suite that runs in about a second without starting containers.

```
python -m venv .venv

# macOS / Linux
source .venv/bin/activate
# Windows PowerShell
.venv\Scripts\Activate.ps1

python -m pip install --upgrade pip
```

Install Airflow matching the Runtime exactly. Airflow requires its constraints file — without it, pip will resolve incompatible transitive dependencies:

```
AIRFLOW_VERSION=2.10.2
PYTHON_VERSION=$(python -c "import sys; print(f'{sys.version_info.major}.{sys.version_info.minor}')" )
CONSTRAINT_URL="https://raw.githubusercontent.com/apache/airflow/constraints-${AIRFLOW_VERSION}/constraints-${PYTHON_VERSION}.txt"

pip install "apache-airflow==${AIRFLOW_VERSION}" \
```

```
apache-airflow-providers-http \
apache-airflow-providers-postgres \
pytest requests psycopg2-binary \
--constraint "${CONSTRAINT_URL}"
```

Windows PowerShell equivalent

```
$AIRFLOW_VERSION="2.10.2"
$PYTHON_VERSION=(python -c "import sys; print(f'{sys.version_info.major}.{sys.version_info.minor}')" )
$CONSTRAINT_URL="https://raw.githubusercontent.com/apache/airflow/constraints-${AIRFLOW_VERSION}/constraints-${PYTHON_VERSION}.txt"
```

```
pip install "apache-airflow==${AIRFLOW_VERSION}" apache-airflow-providers-http
apache-airflow-providers-postgres pytest requests psycopg2-binary --constraint $CONSTRAINT_URL
```

Select the interpreter in VS Code: **Ctrl/Cmd+Shift+P** → **Python: Select Interpreter** → `./.venv/bin/python`.

Quick check that the API contract and transform logic work, with no Docker involved:

```
python include/scripts/smoke_test_pipeline.py
```

Expected tail:

```
[2/3] TRANSFORM
[PASS] row has the expected 6 columns
temperature   : 17.4 C
weathercode    : 3 -> Overcast
[PASS] all range checks passed
```

```
RESULT: all stages passed
```

3. Configure Airflow connections

The DAG references two connection IDs but defines neither. Create them **before** starting Airflow:

```
cp .env.example .env
```

`.env` is git-ignored and is read automatically by `astro dev start`. It defines:

Connection ID	Type	Points at
open_meteo_api	HTTP	https://api.open-meteo.com
postgres_default	Postgres	the weather_db container, database weather

IMPORTANT This is the single most important step in the guide. Airflow ships a built-in connection called `postgres_default` that points at **Airflow's own metadata database**. If you skip this file, the DAG will still appear to succeed — and will write `weather_data` into Airflow's internal database, where `astro dev kill` destroys it. The `.env` entry overrides the built-in, because environment variables take precedence over the metadata database.

Two details in the Postgres connection that trip people up:

- "host": "weather_db" — the **Compose service name**, resolved over the shared Docker network. Not `localhost` (that is the container itself) and not `host.docker.internal` (an unnecessary round trip through the host, and unreliable on Linux).
- "port": 5432 — the **container** port. The host port 5434 is only for connecting from your laptop.

Connections defined by environment variable do **not** appear in the Airflow UI under Admin → Connections. That is expected, not a bug. To manage them in the UI instead, delete them from `.env` and add them through the UI using the same values.

4. Start the stack

```
astro dev start
```

This builds the image and starts five containers: Airflow's webserver, scheduler, triggerer and metadata Postgres, plus the `weather_db` warehouse contributed by `docker-compose.override.yml`.

First run takes several minutes while the Runtime image downloads. Expect:

```
Airflow Webserver: http://localhost:8080
Postgres Database: localhost:5432/postgres
The default Airflow UI credentials are: admin:admin
```

Confirm all five are up:

```
docker ps --format "table {{.Names}}\t{{.Status}}\t{{.Ports}}"
```

You should see a container named `weather_db` publishing `0.0.0.0:5434->5432/tcp`. If it is missing, `docker-compose.override.yml` was not picked up — you are probably not in the project root.

IMPORTANT Port conflicts are the most common startup failure. If 8080 or 5432 are already taken:

```
astro config set webserver.port 8081
astro config set postgres.port 5433
astro dev restart
```

The warehouse uses host port **5434** specifically to stay clear of both.

5. Run the pipeline

- 1 Open `<http://localhost:8080>` and log in with `admin / admin`.
- 2 Find `weather_etl_pipeline` in the DAG list.
- 3 Toggle it **on** (the switch on the left). It is paused by default.
- 4 Click the DAG name, then the **> Trigger DAG** button (top right).

Watch the **Grid** view. Three squares should turn dark green in sequence:

```
extract_weather_data → transform_weather_data → load_weather_data
```

Click any square → **Logs** to see that task's output. In the extract log you will find the resolved request URL; in the load log, the executed SQL.

Running from the command line instead

Test a single task without a full DAG run — useful for tight iteration, as it does not write to the metadata database:

```
astro dev bash # shell into the scheduler container
airflow tasks test weather_etl_pipeline extract_weather_data 2026-07-28
airflow tasks test weather_etl_pipeline transform_weather_data 2026-07-28
airflow tasks test weather_etl_pipeline load_weather_data 2026-07-28
exit
```

`airflow tasks test` runs the task in the foreground and prints logs straight to your terminal.

6. Verify the data landed

Do not trust green squares alone. A task that writes to the wrong database still shows green. Confirm the rows exist where you expect them:

```
docker exec -it weather_db psql -U postgres -d weather -c "SELECT * FROM weather_data ORDER BY
timestamp DESC LIMIT 5;"
```

Expected:

latitude	longitude	temperature	windspeed	winddirection	weathercode	timestamp
51.5074	-0.1278	17.4	11.2	243	3	2026-07-28 07:46:25.028

For a fuller report — row counts, decoded WMO codes, duplicate detection and range assertions:

```
docker exec -i weather_db psql -U postgres -d weather < include/sql/verify_weather_data.sql
```

Querying from VS Code

With the SQLTools extension installed, open the **SQLTools** sidebar icon → **Weather warehouse (local)** → **Connect**. The connection is pre-configured (`localhost:5434`, database `weather`, user/password `postgres`). You can then open `include/sql/verify_weather_data.sql` and run statements with **Ctrl/Cmd+E**.

Proving the whole path independently

To confirm extract, transform *and* load outside Airflow entirely:

```
python include/scripts/smoke_test_pipeline.py --load
```

This performs a real API call and a real insert into `localhost:5434`. If this passes but the DAG fails, the problem is Airflow configuration (connections, networking), not your code or the database.

7. Run the tests

```
pytest tests/ -v
```

IMPORTANT `AIRFLOW_HOME` must point at the project root. `.vscode/settings.json` sets this for VS Code's integrated terminal. In an external terminal, set it yourself:

```
AIRFLOW_HOME=$(pwd) pytest tests/ -v
```

Without it, `dagBag` finds no DAGs, every parametrised test is skipped, and the suite reports a **misleading pass**.

Two suites are present:

`tests/dags/test_weather_etl.py` — 9 unit tests written for this pipeline. They execute the real task callables with hooks mocked out, replaying the actual load SQL against in-memory SQLite. No Docker, no network, under a second:

```
test_dag_has_three_tasks_in_a_linear_chain PASSED
test_dag_schedule_is_daily_without_catchup PASSED
test_extract_builds_the_expected_endpoint PASSED
test_extract_raises_on_non_200 PASSED
test_transform_flattens_to_the_warehouse_row_shape PASSED
test_transform_echoes_requested_coordinates_not_api_coordinates PASSED
test_transform_fails_loudly_when_current_weather_is_absent PASSED
test_load_creates_table_and_inserts_row PASSED
test_load_is_append_only_so_reruns_duplicate_rows PASSED
```

`tests/dags/test_dag_example.py` — Astronomer's stock integrity checks. **These currently fail for `weather_etl_pipeline`, and that is a genuine finding, not a broken test:**

```
FAILED test_dag_tags[dags/etlweather.py] - AssertionError: ... has no tags
FAILED test_dag_retries[dags/etlweather.py] - TypeError: '>=' not supported between 'NoneType' and 'int'
```

To make both pass, edit `dags/etlweather.py` (see Case Study §6.1):

```
default_args = {
    'owner': 'airflow',
    'start_date': days_ago(1),
    'retries': 3,
    'retry_delay': timedelta(minutes=5),
}

with DAG(dag_id='weather_etl_pipeline',
        default_args=default_args,
        schedule_interval='@daily',
        catchup=False,
        tags=['etl', 'weather', 'open-meteo']) as dags:
```

with `from datetime import timedelta` added at the top. Re-run `pytest tests/ -v` — all should pass.

8. Everyday commands

Task	Command
Start	<code>astro dev start</code>
Stop (keep data)	<code>astro dev stop</code>
Restart after changing <code>.env</code> or Compose	<code>astro dev restart</code>
Destroy everything including the metadata DB	<code>astro dev kill</code>
Shell into the scheduler	<code>astro dev bash</code>
Follow scheduler logs	<code>astro dev logs --scheduler --follow</code>
List parsed DAGs	<code>astro dev bash -c "airflow dags list"</code>

Task	Command
Show import errors	<code>astro dev bash -c "airflow dags list-import-errors"</code>
Test one task	<code>astro dev bash -c "airflow tasks test weather_etl_pipeline extract_weather_data 2026-07-28"</code>
psql into the warehouse	<code>docker exec -it weather_db psql -U postgres -d weather</code>
Run tests	<code>AIRFLOW_HOME=\$(pwd) pytest tests/ -v</code>

Changes to files in `dags/` are picked up automatically within about 30 seconds — no restart needed. Changes to `.env`, `requirements.txt`, `Dockerfile` or `Compose` files **do** require `astro dev restart`.

9. Troubleshooting

The `conn_id 'open_meteo_api'` isn't defined

The connection is missing. Confirm `.env` exists (not just `.env.example`) and restart:

```
ls -la .env && astro dev restart
```

Verify it reached the container:

```
astro dev bash -c "env | grep AIRFLOW_CONN"
```

The DAG succeeds but `weather_data` is empty

Almost certainly the `postgres_default` collision (Case Study §6.3) — the rows went into Airflow's metadata database. Check:

```
docker exec -it postgres psql -U postgres -d postgres -c "\dt"
```

If `weather_data` appears there, your `.env` override was not applied. Confirm the container sees it, then `astro dev restart`.

could not translate host name "weather_db"

The warehouse container is not on the Airflow network. Confirm it is running and attached:

```
docker ps --filter name=weather_db
docker network inspect $(docker network ls --filter name=airflow -q) --format '{{range .Containers}}{{.Name}} {{end}}'
```

`weather_db` must appear in that list. If not, ensure `docker-compose.override.yml` is at the project root and run `astro dev restart`.

Ports are not available / address already in use

Something already owns 8080 or 5432:

```
astro config set webserver.port 8081
astro config set postgres.port 5433
astro dev restart
```

Broken DAG banner in the UI

```
astro dev bash -c "airflow dags list-import-errors"
```

Note that a `RemovedInAirflow3Warning` about `days_ago` or `schedule_interval` is a **warning**, not an import error — the DAG still loads on Airflow 2.10.2.

Task fails with `Failed to fetch weather data: 429`

Open-Meteo rate-limits its free tier. Wait a minute and re-run. This is exactly the scenario retries would absorb — see Case Study §6.1.

Tests all skip and report a pass

`AIRFLOW_HOME` is unset. Use `AIRFLOW_HOME=$(pwd) pytest tests/ -v`.

Containers keep restarting or the scheduler is slow

Docker Desktop is under-provisioned. Raise memory to 4 GB or more in Settings → Resources, then `astro dev restart`.

Reset the warehouse to an empty state

```
astro dev stop
docker volume rm etlweather_weather_db_data
astro dev start
```

The volume name is prefixed with the project directory name; run `docker volume ls` to confirm.

10. Suggested exercises

Ordered by difficulty. Each corresponds to a finding in the Case Study.

- 1 **Add retries and tags** (§6.1) — make the stock test suite pass. Trivial, high value.
- 2 **Modernise the deprecated APIs** (§6.5) — swap `days_ago(1)` for `pendulum.datetime(...)` and `schedule_interval=` for `schedule=`. Confirm the warnings disappear.
- 3 **Preserve observation time** (§4.4) — carry `current_weather['time']` through the transform into a new `observation_time` column, so you record when the weather *happened*, not when you wrote the row.
- 4 **Make the load idempotent** (§6.2) — add a primary key on `(latitude, longitude, observation_time)` and switch to `INSERT ... ON CONFLICT ... DO UPDATE`. Prove it by inverting `test_load_is_append_only_so_reruns_duplicate_rows` to assert a count of **1**.
- 5 **Add a data-quality gate** (§6.6) — insert a validation task between transform and load that raises on out-of-range values. Encode the same rules as `include/sql/verify_weather_data.sql`.
- 6 **Support multiple cities** (§6.7) — convert to dynamic task mapping with `.expand()` and add a `city` column. One DAG, N parallel extracts.
- 7 **Decode weather codes at load time** — join to a `wmo_codes` lookup table so analysts read "Overcast" instead of `3`.

- 8 **Add alerting** — configure `on_failure_callback` to post to Slack or email, so failures reach you without watching the UI.